Risk Factors and Characteristics of Intraoperative Seizures During Awake Craniotomy: A Retrospective Cohort Study of 562 Consecutive Patients With a Space-occupying Brain Lesion

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Introduction: Intraoperative seizures (IOSs) during awake craniotomy (AC) are associated with significant morbidity. The reported incidence of IOS is between 3% and 30%. The aim of this study was to identify risk factors for IOS during AC for elective resection or biopsy of a space-occupying brain lesion.

Methods: In this retrospective study, we reviewed the records of all awake craniotomies performed by a single neurosurgeon at a single university hospital between July 2006 and December 2018. IOS was defined as a clinically apparent seizure that occurred in the operating room and was documented in the medical records. Explanatory variables were chosen based on previously published literature on risk factors for IOS.

Results: Five hundred and sixty-two patients had a total of 607 AC procedures during the study period; 581 cases with complete anesthesia records were included in analysis. Twenty-nine (5.0%) IOS events were reported during 29 (5%) awake craniotomies. Most seizures (27/29; 93%) were focal in nature and did not limit planned intraoperative stimulation mapping. Variables associated with IOS at a univariate P-value <0.1 (frontal location of tumor, preoperative radiotherapy, preoperative use of antiepileptic drugs, intraoperative use of dexmedetomidine, and intraoperative stimulation mapping) were included in a multivariable logistic regression. Frontal location of tumor (adjusted odds ratio: 5.68, 95% confidence interval: 2.11-15.30) and intraoperative dexmedetomidine use (adjusted odds ratio: 2.724, 95% confidence interval: 1.24-6.00) were independently associated with IOS in the multivariable analysis.

Conclusions: This study identified a low incidence (5%) of IOS during AC. The association between dexmedetomidine and IOS should be further studied in randomized trials as this is a modifiable risk factor.

Key Words: awake craniotomy, brain neoplasms, dexmedetomidine, seizures

(*J Neurosurg Anesthesiol* 2023;35:194–200)

Awake craniotomy (AC) with intraoperative stimulation mapping is commonly indicated for the surgical resection of brain tumors adjacent to eloquent cortex.[1] AC is associated with a greater extent of tumor resection,[2,3] fewer postoperative complications,[4] fewer neurological deficits,[2,5] and extended survival[2,4] when compared with craniotomy with general anesthesia. In some centers, AC is also used for tumors involving noneloquent brain regions in order to avoid general anesthesia, maximize tumor resection, and facilitate early discharge from hospital.[5–7] Intraoperative seizures (IOSs) are one of the worrisome complications during AC with a reported incidence between 3% and 30%.[1] Most seizures are associated with direct electrical stimulation of the cortex during intraoperative stimulation mapping.[8] IOSs may lead to loss of the patient's cooperation, airway complications, and unplanned conversion to general anesthesia.

Anesthesia for AC can be managed with monitored anesthesia care (MAC) with conscious sedation or an asleep-awake-asleep (SAS) technique.[1,9] Several risk factors for IOS during AC have been reported in the literature. These include younger age,[10,11] frontal lobe involvement,[11–14] preoperative seizure history,[11,15] use of multiple antiepileptic drugs (AEDs),[11] low-grade glioma,[11] positive mapping,[16]

Received for publication May 13, 2021; accepted July 23, 2021.
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Supported by the Department of Anesthesia, Toronto Western Hospital, University of Toronto.
Preliminary results from this study have been presented at the following conferences: Canadian Anesthesiologists' Society (CAS) Virtual Annual Meeting, June 19-22, 2020. Society for Neuroscience in Anesthesiology and Critical Care (SNACC) Virtual Annual Meeting, September 11-13, 2020.
The authors have no conflicts of interest to disclose.
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DOI: 10.1097/ANA.0000000000000798

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tumor volume,[16] *isocitrate dehydrogenase-1 (IDH1)* gene mutation[12] and certain electrical stimulation parameters including longer stimulus duration and 50 to 60 Hz bipolar stimulation.[17] In a recent meta-analysis, MAC was associated with a higher incidence of seizures but a lower failure rate compared with the SAS technique; longer term outcomes such as survival or cancer recurrence were not assessed in this meta-analysis.[18]

The influence of specific anesthesia drugs on seizure incidence during MAC for AC is poorly defined. Therefore, this study aims to define the incidence, characteristics, and management of IOS during AC for brain lesions. In addition, we also explore the various anesthetic and surgical risk factors for IOS in this patient cohort.

METHODS

After approval from the Institutional Ethics Review Board (#18-6131 and #21-5410) including waiver of informed patient consent, we conducted a retrospective chart review to collect surgical and anesthetic data for all patients over 18 years of age who underwent AC for resection or biopsy of a space-occupying brain lesion by a single neurosurgeon (M.B.) at our institution between 13 July 2006 and 31 December 2018. The AC database of this neurosurgeon was maintained prospectively. In a previous publication, Takami et al[19] used the demographic and surgical data in this database to analyze the risk factors for intraoperative complications leading to deepening of sedation, which made the surgery no longer an awake procedure. This adverse event occurred in 21 patients, 4 of whom had seizures; no anesthesia related data were reported in that study. In the current study, we queried the completed database after subsequent inclusion of anesthetic data to identify risk factors for IOS. In addition to the anesthetic data, we included relevant demographic, surgical and pathologic data reported by Takami et al.[19] We excluded patients with missing anesthesia records in the electronic patient record. Follow-up was limited to the day of surgery.

Anesthetic Management

Standard anesthesia practice for AC at our institution is MAC with conscious sedation. All patients scheduled for AC are evaluated in the preoperative clinic by an anesthesiologist to assess fitness for AC and obtain informed consent. AEDs are continued until surgery if applicable, but it is not routine practice to administer prophylactic preoperative or intraoperative AEDs in our institution. Canadian Anesthesiology Society standard monitoring is used for all cases, while invasive hemodynamic monitoring (arterial line) is rarely used. Supplemental oxygen is delivered by nasal prongs with capnography monitoring. Invasive airway management is used only if mandated to manage an intraoperative complication. A combination of intravenous sedatives (midazolam and/or propofol and/or dexmedetomidine) and opioids (fentanyl and/or remifentanil) is administered for maintaining conscious sedation at the discretion of the anesthesiologist. Deep sedation is usually achieved with a propofol bolus for application of the Sugita head clamp. The patient is then allowed to emerge from sedation to allow optimal positioning on the operating table. Sedation is resumed for the craniotomy and dural incision. Conscious sedation is maintained primarily following one of two protocols: either with a propofol infusion (with or without remifentanil) or a dexmedetomidine infusion, which can be combined with a propofol infusion during the more stimulating parts of the procedure. All sedatives are stopped 15 to 20 minutes before intraoperative mapping, if applicable, and restarted after completion of mapping. IOSs are managed by cessation of electrical cortical stimulation if applicable, irrigation of the cortex with cold saline,[20] and/or treatment with intravenous midazolam, propofol or AEDs. Postoperatively, patients are observed in the postanesthesia care unit and then discharged to either the day surgery unit, or the neurosurgical ward. The main change in anesthesia management during the study period was the introduction of dexmedetomidine into our practice in April 2012. Other aspects of anesthesia management remained unchanged throughout the study.

Surgical Procedure

The surgical approach used by the neurosurgeon (M.B.) in this study has been described previously.[6,19,21,22] An AC with intraoperative stimulation mapping is performed for brain lesions adjacent to eloquent cortex according to preoperative imaging. AC without intraoperative mapping is used with lesions remote from eloquent regions to avoid general anesthesia and accelerate patient recovery. After initiation of conscious sedation, local anesthesia is administered as a circumferential field block over the surgical incision and at the pin sites; the head is then fixed in a Sugita head clamp. Image-guided frameless navigation is used in all cases. Intraoperative stimulation mapping is performed using a bipolar stimulator to identify eloquent cortical and subcortical structures. The initial stimulation intensity is set at 8 mA for all patients and increased to 10 or 12 mA if no eloquent structures are identified and higher intensity stimulation is deemed safe by the neurosurgeon. Electrocorticography is not used during functional mapping during AC at our institution. Motor, sensory, and/or language functions are evaluated intraoperatively with standardized tests. Eloquent cortical and subcortical areas identified by positive mapping are avoided during tumor resection. This surgeon has used a similar approach since 1998 for patient selection and surgical technique, and it was not modified during the study period.

Data Sources, Measurements, and Definitions

The data sources were the neurosurgeon's prospective database and the institutional electronic patient record (QuadraMed Corporation, Reston, VA). Documents reviewed included preoperative anesthesia consultations, anesthesia records, postanesthesia care unit nursing records, clinical notes, medication history, radiologic records, pathology reports, surgical dictation notes, and clinic visit

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notes. Collection of surgical and anesthetic data was performed by 2 independent groups (H.T. and M.B. for surgical data and G.P.-L., S.S., K.W.L., and A.D. for anesthetic data) and compiled in Excel (Microsoft, Redmond, WA). Any discrepancy between surgical and anesthetic databases was resolved by consulting the electronic record for validation of data collected (G.P.-L.) and discussion with the senior author (L.V.).

Explanatory variables were chosen based on previously published literature and clinical experience. We included age, sex, Karnofsky performance score, tumor location, tumor volume, side of surgery, first or repeat surgery, preoperative history of seizures, preoperative use of AEDs (before and continued on the day of surgery), preoperative use of benzodiazepines, length of surgery, intraoperative stimulation mapping, and intraoperative medications used. *IDH1* mutation and propofol infusion doses were not included in the analysis because these data were missing from more than 50% of patient records. Descriptive variables relative to IOS events included timing of seizure during the surgical procedure, type of seizure, treatments used, and intraoperative complications relative to the seizure.

An IOS was defined as a clinically apparent seizure that occurred in the operating room and was documented in either the surgical or anesthesiology records. A preoperative history of seizure was defined as any event described as a potential seizure in the preoperative medical, surgical or anesthetic records, or leading to initiation of an antiseizure medication. Preoperative use of AEDs was classified as either "none," "single AED," or "multiple AEDs"; these excluded gabapentinoids used for pain management and benzodiazepines other than clobazam. Location of tumor was defined as the main cerebral lobe involved by the lesion operated according to radiologic and surgical records.

Statistical Analysis

Sample size was determined by the number of cases available during the study period; no formal sample size calculation was performed before the study. Descriptive data are presented as number (%) for categorical variables, mean and SD for parametric variables, and median with interquartile range for nonparametric variables. Normality of continuous data was assessed by visual inspection of the histogram and Kolmogorov-Smirnov tests. The incidence of IOS is presented for the study population, and for the periods preceding and following introduction of dexmedetomidine to our practice.

In univariate analysis, χ^2 or the Fischer exact tests (when any expected value was under 5) were used for categorical variables, *t* tests for parametric variables and Mann-Whitney *U* tests for nonparametric variables. Categories with values of 0 or 1 in either group were collapsed with another category in the univariate analysis; for location of tumor all categories other than "frontal" were collapsed, and for side of surgery "bilateral" was collapsed with "left."

To assess the interaction of confounders with significant variables identified in univariate analysis, a

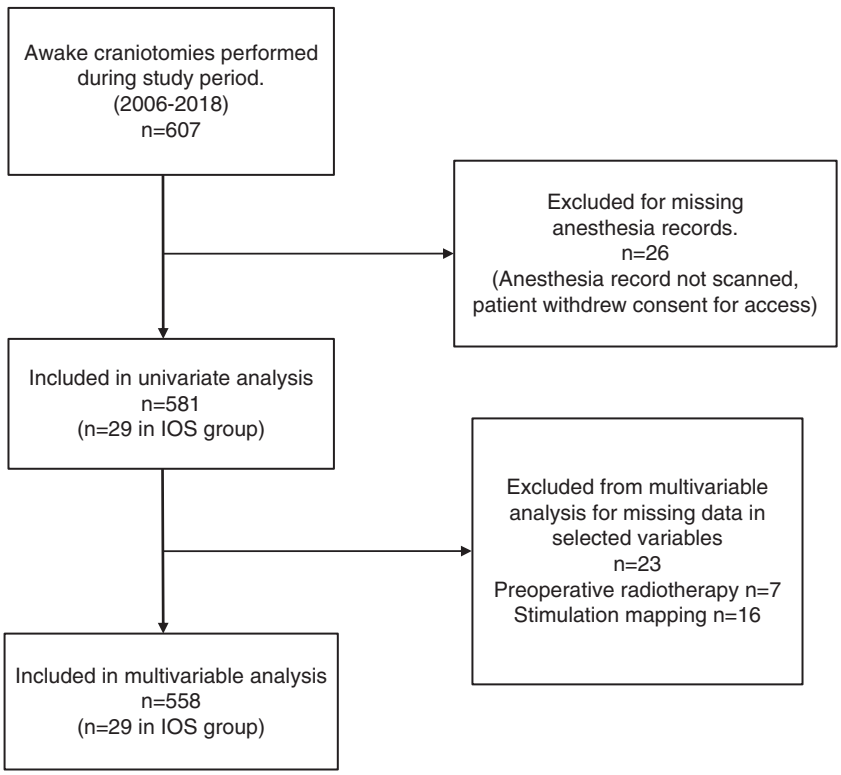


FIGURE 1. Study flow chart. IOS indicates intraoperative seizure.

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multivariable binomial logistic regression was performed. We included in the model the variables associated with seizures at a univariate P-value < 0.1 using a forced-entry method. Preoperative use of AEDs was not included in the multivariable model because of a significant association ($P < 0.001$) with preoperative history of seizures, which is an important potential risk factor for IOS. An alternate multivariable analysis replacing preoperative history of seizures with preoperative use of AEDs has been performed as a sensitivity analysis. Missing variables were managed by pairwise deletion in the univariate analysis and listwise deletion of cases in the multivariable analysis. There was no preplanned subgroup analysis. Statistical analyses were performed using SPSS and a 2-sided P-value <0.05 was considered statistically significant.

RESULTS

Six hundred and seven ACs were performed in 562 patients during the study period, as reported previously.[19] Two bilateral ACs were each considered to be a single case, and 26 cases were excluded from the analysis because of missing anesthesia records (Fig. 1). Five hundred eighty-one cases were included in the univariate analysis. Twenty-nine of these 581 cases had an IOS, representing an IOS incidence of 5% in our study population. The incidence of IOS increased from 3.3% to 7.0% following the introduction of dexmedetomidine into our practice (χ^2 $P = 0.045$). Eighteen of 256 patients had an IOS after dexmedetomidine was introduced, 13 of whom (72%) received dexmedetomidine intraoperatively.

TABLE 1. Univariate Analyses of Risk Factors for Intraoperative Seizures: Patient Characteristics

	IOS (n = 29)	No-IOS (n = 552)	P
Age (y) (mean ± SD)	52.1 ± 13.0	55.9 ± 15.7	0.20
Female sex, n (%)	15 (51.7)	252 (45.7)	0.52
KPS (mean ± SD)	73.1 ± 14.2	77.2 ± 13.8	0.12
Side of surgery, n (%)			
Left	15 (51.7)	309 (56.0)	0.63
Right	14 (48.3)	241 (43.7)	
Bilateral	0 (0)	2 (0.4)	
Repeat surgery, n (%)	4 (13.8)	101 (18.3)	0.54
Location of tumor, n (%)			**< 0.001**
Frontal	24 (82.8)	251 (45.5)	
Parietal	4 (13.8)	121 (21.9)	
Temporal/insular	1 (3.4)	126 (22.8)	
Occipital	0 (0)	16 (2.9)	
Other	0 (0)	34 (6.2)	
Multiple	0 (0)	4 (0.7)	
Pathology, n (%)			0.19
Low-grade glioma	4 (13.8)	68 (12.3)	
High-grade glioma	10 (34.5)	290 (52.5)	
Metastasis/radiation necrosis	12 (41.4)	159 (28.8)	
Other	3 (10.3)	35 (6.3)	
Tumor volume in cm^3, median (IQR)	23.3 (11.0-36.5)	20.7 (8.6-39.1)	0.65
Preoperative chemotherapy, n (%)	11 (39.3)	160 (29.8)	0.29
Preoperative radiotherapy, n (%)	11 (37.9)	117 (21.5)	**0.04**
Preoperative history of seizure, n (%)	18 (62.1)	234 (42.4)	**0.04**
Number of AEDs, n (%)			**0.04**
2 or more AEDs	6 (20.7)	47 (8.5)	
One AED	11 (37.9)	171 (31.0)	
None	12 (41.4)	334 (60.5)	
Preoperative benzodiazepines, n (%)	2 (6.9)	53 (9.6)	1.00

Values in bold indicate statistically significant differences ($P < 0.05$).
AEDs indicates antiepileptic drugs; IOS, intraoperative seizures; IQR, interquartile range; KPS, Karnofsky Performance Score.

TABLE 2. Univariate Analyses of Risk Factors for Intraoperative Seizures: Intraoperative Data

	IOS (n = 29)	No-IOS (n = 552)	P
Type of surgery, n (%)			0.38
Tumor resection	25 (86.2)	439 (79.5)	
Biopsy or aspiration	4 (13.8)	113 (20.5)	
Length of surgery in min, mean (SD)	117.7 (36.7)	111,3 (38.1)	0.38
Intraoperative stimulation mapping, n (%)	25 (86.2)	382 (71.3)	0.08
Intraoperative steroids, n (%)	26 (89.7)	484 (87.7)	1.00
Intraoperative midazolam (before seizure), n (%)	19 (65.5)	428 (77.5)	0.13
Intraoperative remifentanil, n (%)	14 (48.3)	290 (52.5)	0.65
Intraoperative dexmedetomidine, n (%)	13 (44.8)	124 (22.5)	**0.006**
Intraoperative prophylactic phenytoin, n (%)	0 (0)	4 (0.7)	1.00
Intraoperative propofol infusion, n (%)	24 (82.8)	474 (85.9)	0.59

Values in bold indicate statistically significant differences ($P < 0.05$).
IOS indicates intraoperative seizures.

The preoperative and intraoperative characteristics for each group are presented in Tables 1 and 2. 43.3% of patients had any preoperative history of seizures, and 40.4% were on AEDs. In the seizure group, 25 patients had tumor resection surgery and 4 patients had tumor biopsy or aspiration procedures. Twenty-five of 29 (86%) patients with IOS underwent intraoperative mapping, including the 4 patients who had tumor aspiration or biopsy.

The characteristics of IOS events are shown in Table 3. Most seizures were focal in nature (93.1%) and were induced by electrical stimulation (69.0%). Nine patients had a seizure unrelated to intraoperative stimulation mapping, of whom 6 (66.7%) had a previous seizure history. Most seizures (62.1%) were treated with intravenous medication, while others (37.9%) resolved spontaneously or following irrigation of the cortex with cold saline. No patient required conversion to general anesthesia because of an IOS. One patient who had an isolated stimulation-induced seizure later required general anesthesia because of intraoperative air embolism. Planned mapping could be performed in most patients (26/29; 89.7%) with IOS.

Regarding the risk factors for IOSs, univariate analysis revealed a significant association between IOS and frontal tumor location (82.8% vs. 45.5%; $P < 0.001$), preoperative radiotherapy (37.9 vs. 21.5%; $P = 0.0380$), preoperative

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TABLE 3. Seizure Characteristics, Treatment and Complications (n = 29)

	N (%)
Type of seizure	
Focal	27 (93.1)
Generalized	2 (6.9)
Timing of seizure	
During head pinning	2 (6.9)
During surgical approach	2 (6.9)
Stimulation-induced seizure	20 (69.0)
During resection or biopsy	4 (13.8)
Unclear timing in record	1 (3.4)
Treatment of seizure	
Spontaneous resolution	7 (24.1)
Cold saline alone	4 (13.8)
Propofol bolus alone	7 (24.1)
Cold saline+propofol	8 (27.6)
Cold saline+benzodiazepine	1 (3.4)
Propofol+benzodiazepine	2 (6.9%)
Intraoperative use of antiepileptic drugs (AEDs)	
Therapeutic phenytoin	4 (13.8)
No intraoperative AED	25 (86.2)
Intraoperative complications related to seizure	
Planned mapping not performed	3 (10.3)
Conversion to general anesthesia	0

history of seizures (62.1 vs. 42.4%; $P = 0.0371$), preoperative AED use (2 or more AEDs 20.7% vs. 8.5%; 1 AED 37.9% vs. 31.0%; $P = 0.0378$) and intraoperative dexmedetomidine use (44.8% vs. 22.5%; $P = 0.006$).

The following variables were entered in the multivariable model (Table 4), based on a P-value <0.1 in univariate analysis: frontal location of tumor, preoperative radiotherapy, preoperative use of AEDs (classified as none, single AED, or multiple AEDs), intraoperative dexmedetomidine use, and intraoperative stimulation mapping. Frontal location of tumor (adjusted odds ratio: 5.681, 95% confidence interval: 2.11-15.30) and intraoperative use of dexmedetomidine (adjusted odds ratio: 2.724, 95% confidence interval: 1.24-6.00) were independently associated with IOS.

In a post hoc exploratory analysis, we compared the use of propofol infusion between patients who did and did not receive dexmedetomidine to test the hypothesis that propofol use would be lower with dexmedetomidine use. Propofol infusion use was significantly lower in the dexmedetomidine group; 77 of 137 patients (56.2%) in the dexmedetomidine group received a propofol infusion compared with 421 of 444 patients (94.8%) who did not receive dexmedetomidine (χ^2 $P < 0.001$). However, individual propofol doses cannot be reported as these were missing from the paper records before the introduction of electronic records.

TABLE 4. Results From Multivariate Analysis of Risk Factors for Intraoperative Seizure

	aOR	95% CI	*P*
Frontal location of tumor	**5.681**	**(2.11-15.30)**	**0.001**
Preoperative radiotherapy	1.974	(0.87-4.49)	0.11
Preoperative history of seizures	1.539	(0.68-3.46)	0.30
Intraoperative use of dexmedetomidine	2.724	(1.24-6.00)	**0.01**
Intraoperative stimulation mapping	2.886	(0.96-8.70)	0.06

Values in bold indicate statistically significant differences ($P < 0.05$).
AEDs indicates antiepileptic drugs; aOR, adjusted odds ratio; CI, confidence interval.

DISCUSSION

In this study, we investigated the incidence and risk factors for IOS in a large series of consecutive patients undergoing AC for brain tumor, operated by a single neurosurgeon. The incidence of IOS was 5% in our study population. Most seizures were focal in nature and caused by electrical stimulation mapping, which is in line with previously published data.[15] No patients required conversion to general anesthesia because of IOS, and planned intraoperative mapping could be achieved in the majority (89.7%) of patients with IOS.

Our study showed that among various factors, frontal location of the tumor and intraoperative use of dexmedetomidine were independently associated with IOS. Frontal tumor location as an independent risk factor for IOS has been reported previously.[11–13] In a recent literature review of stimulation-induced IOS during AC, the frontal location of the tumor was the risk factor identified in most studies (5 of 7).[14] This risk of IOS with frontal lesions is probably related to electrical stimulation of the motor cortex, premotor cortex, and supplemental motor area, which may unsurprisingly lead to motor seizures.

The association we found between dexmedetomidine and IOS during AC with conscious sedation has not been previously reported. Dexmedetomidine has been successfully used in AC for almost 20 years because of its favorable pharmacokinetic and pharmacodynamic characteristics.[23–25] Its benefits for AC include maintenance of spontaneous breathing and patient cooperation,[26] reduction of respiratory adverse events[24] and shorter arousal time.[27] There is no clear evidence linking dexmedetomidine and seizures in the published human literature. Dexmedetomidine has been shown to reduce seizure thresholds in some animal models using very large doses (over 10 μg/kg) that are not encountered in clinical practice.[28,29] However, more recent animal studies have demonstrated anticonvulsant effects of dexmedetomidine.[30–32] In patients undergoing epilepsy surgery, dexmedetomidine has shown to have minimal effect on electrocorticography making it a suitable agent for epilepsy surgery.[33,34] One small retrospective study on AC reported significantly more seizures with MAC using dexmedetomidine compared with an SAS technique.[35] In a meta-analysis of recent studies comparing MAC to SAS for AC, the incidence of IOS was higher in the MAC group which included three studies that used dexmedetomidine-based conscious sedation.[18] However, the influence of dexmedetomidine on seizures in these reports is not conclusive given the comparison of dexmedetomidine sedation with an SAS technique in which a lower seizure rate would be expected because of deeper anesthesia. Other large studies investigating IOS during AC have not reported on the influence of specific anesthesia regimens.[11,14] In contrast, randomized controlled trials

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comparing propofol with dexmedetomidine-based conscious sedation have either not reported IOS[27] or not found significant differences in the incidence of IOS.[24]

In our study, the association between dexmedetomidine and IOS should be viewed as hypothesis generating, without implying a causal relationship between dexmedetomidine use and IOS. For instance, dexmedetomidine-based sedation in AC generally implies lower use of other anesthetic agents, such as propofol, that are known to raise the seizure thresholds. Indeed, in our cohort propofol infusion was used less frequently in patients who received dexmedetomidine compared with those receiving a dexmedetomidine-free regimen; however, use of propofol infusion was not significantly different between the IOS and no-IOS groups in our study. The association of dexmedetomidine with seizures could, therefore, be explained by less frequent use of anesthetic medication with GABAergic activity for conscious sedation when dexmedetomidine is used as the primary agent. The shallower depth of anesthesia provided by dexmedetomidine could also explain some IOS events. For example, in a report on AC without any sedation,[36] eight patients of on 50 (16%) experienced IOS. This is a higher rate of seizures than other recent AC series,[14,37] suggesting that patients who are more awake may have an increased risk of IOS during AC. In summary, though dexmedetomidine may be associated with a slightly elevated risk of IOS during AC it may not be the direct cause of the seizures.

We also found other associations in the univariate analysis that were not significant in our multivariable model: preoperative seizure history and AED use which have been reported previously,[14] and preoperative radiotherapy. Preoperative radiotherapy could be a marker of metastatic or recurrent disease, which may mediate the increased seizure risk. The loss of significance in the multivariable analysis for these variables could be the result of insufficient power in our study cohort because of the low event rate, or interaction with independent risk factors or other unmeasured factors. In any case, it may still be clinically useful to suspect a higher IOS risk in patients with a previous seizure history or preoperative radiotherapy, especially when these are combined with other risk factors.

This study has several limitations. First, because of its retrospective design there is risk of missing, incomplete or inconsistent data in the available records. Since we relied on clinical detection and documentation of seizures by the neurosurgeon and/or anesthesiologist, not all seizure events may have been detected. Some potentially interesting data such as propofol dose, *IDH1* mutation status or cortical stimulation parameters could not be included in the analysis because of a high proportion of missing data. Second, though this is a single neurosurgeon cohort, multiple anesthesiologists, fellows and residents were involved in providing patient care over more than a decade. While the overall surgical approach and patient selection remained consistent, there may have been variations in techniques or skills over the years. Because choice of anesthesia drugs was at the discretion of the anesthesiologist, the influence of unmeasured confounding factors or allocation bias cannot be excluded. The incidence of IOS was higher after the introduction of dexmedetomidine for AC, a finding probably related to this change in practice but which may also be attributable to other, unmeasured, confounding factors. Third, the relatively low number of seizure events limited the number of variables allowed in multivariable analysis and the statistical power to detect significant associations. Screening of variables with univariate analyses is a common approach for selection of variables to enter in a multivariable model when searching for risk factors for rare outcomes, but may lead to model overfitting.[38] Our model included 5.8 events per variable; this is just above the 2 to 4 events per variable range where excessive overfitting is more problematic but below the traditionally recommended 10 events per variable.[39] This approach also led us to exclude interesting variables such as age or pathology from our multivariable model because of weaker associations with the outcome variable; this limited our ability to account for all potential confounding factors. Also, the low number of events could explain why seemingly important risk factors for IOS such as intraoperative stimulation mapping did not reach statistical significance in our analyses. Fourth, the amount of variance explained by the multivariable model was relatively low (16.2%). Because of the low number of events, the goal of this study was not to construct a fully predictive IOS multivariable model but to screen for IOS risk factors while controlling for confounding from other variables associated with the outcome; therefore, a low explanatory power of the model may be expected. Thus, our results should be viewed as hypothesis generating until replicated in a prospective setting. Finally, the disadvantage of our single center study in an academic hospital with a high volume of AC procedures is its limited external validity. Our findings could be dependent on factors specific to our institutional practice and may not be generalizable to other settings. Notably, the performance of AC without intraoperative stimulation monitoring to accelerate recovery and limit exposure to general anesthesia, although advocated by some authors,[5,40] is not widespread practice at the moment, and may reduce the incidence of IOS compared with other cohorts. Some risk factors for IOS may also be different in patients having nonstimulation-related seizures. For example, though the underlying seizure risk may play a more important role in this subpopulation with a higher prevalence of preoperative seizures, events were insufficient to perform meaningful subgroup analyses to test this hypothesis.

In conclusion, our study reported a low incidence (5%) of IOS during AC. Most seizures were focal in nature and did not lead to AC failure. Of all selected variables, an association between intraoperative dexmedetomidine use and IOS was noted. This study is the first to report this association in a population of patients undergoing AC with conscious sedation. Because of the inadequate sample size limiting our ability to construct a multivariable model including all potential confounders, it is difficult to extrapolate this finding. The association between dexmedetomidine and IOS should be further investigated in randomized trials as this is a modifiable risk factor.

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